CSharPN

A Code-First Framework for Coloured Petri Net Modelling with C# as Inscription Language

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Underneath it, the older gap between **academia and industry**.

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CSharPN — a framework for working with Coloured Petri Nets in C#, with a VS Code extension for visualising and executing models.

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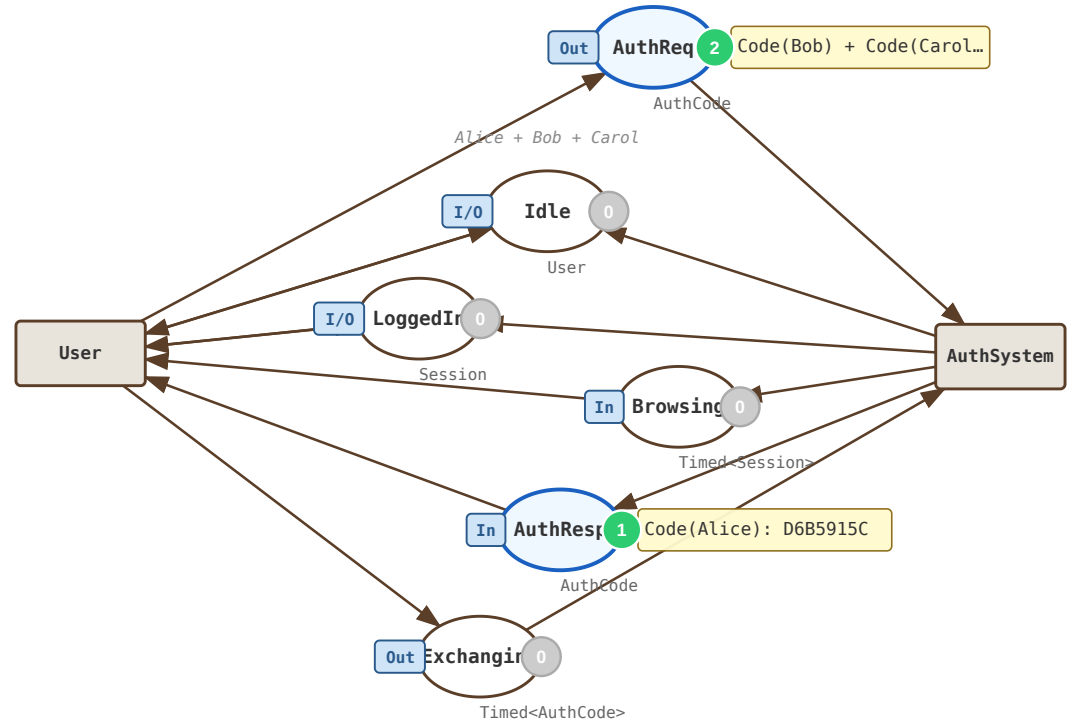
> Simulation-based development — with concurrency, resource sharing, timing, and hierarchy modelled *explicitly*.

Running example:

OAuth2 authorisation code flow

Exercises all four features:

Top page: shared places link the two substitution transitions.



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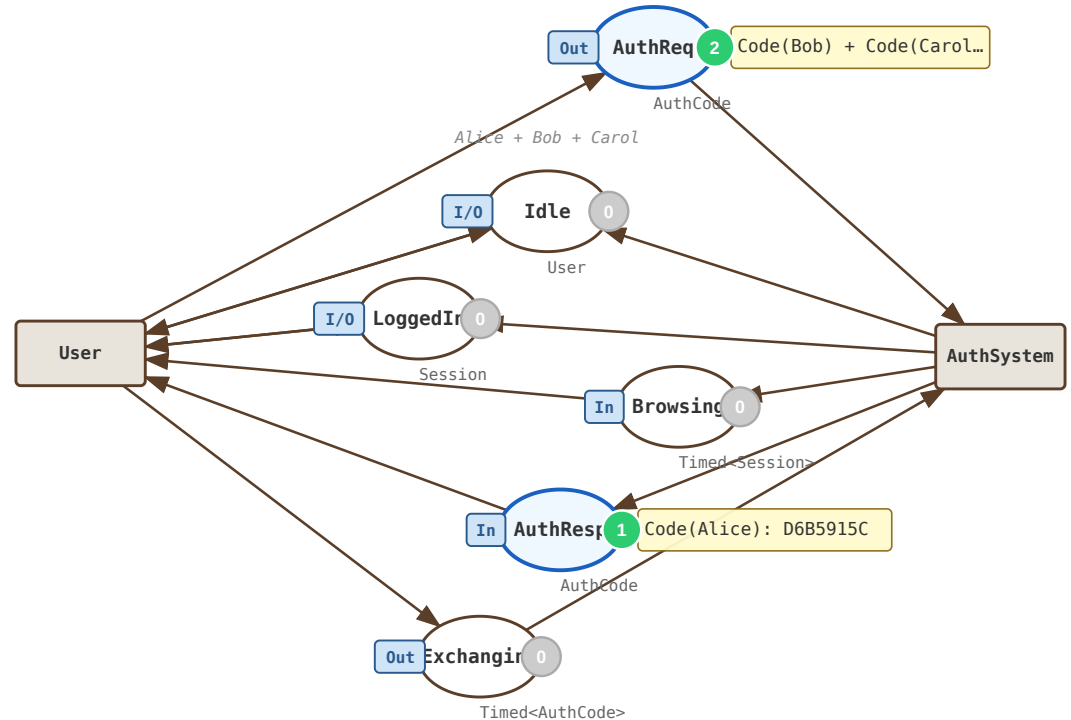

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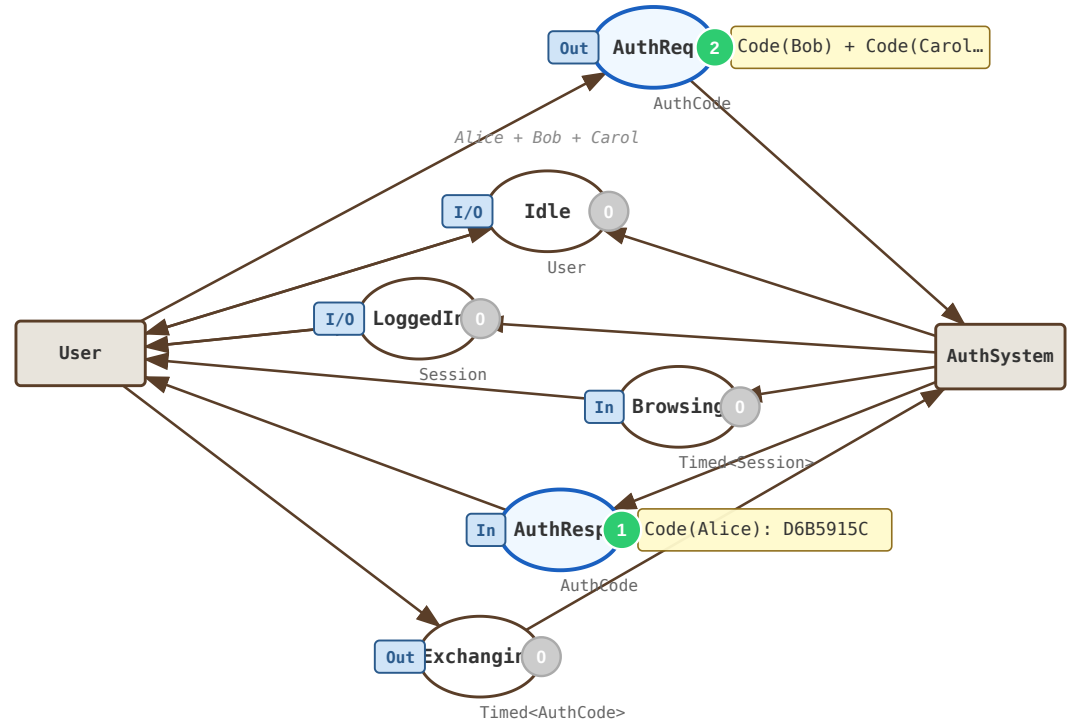

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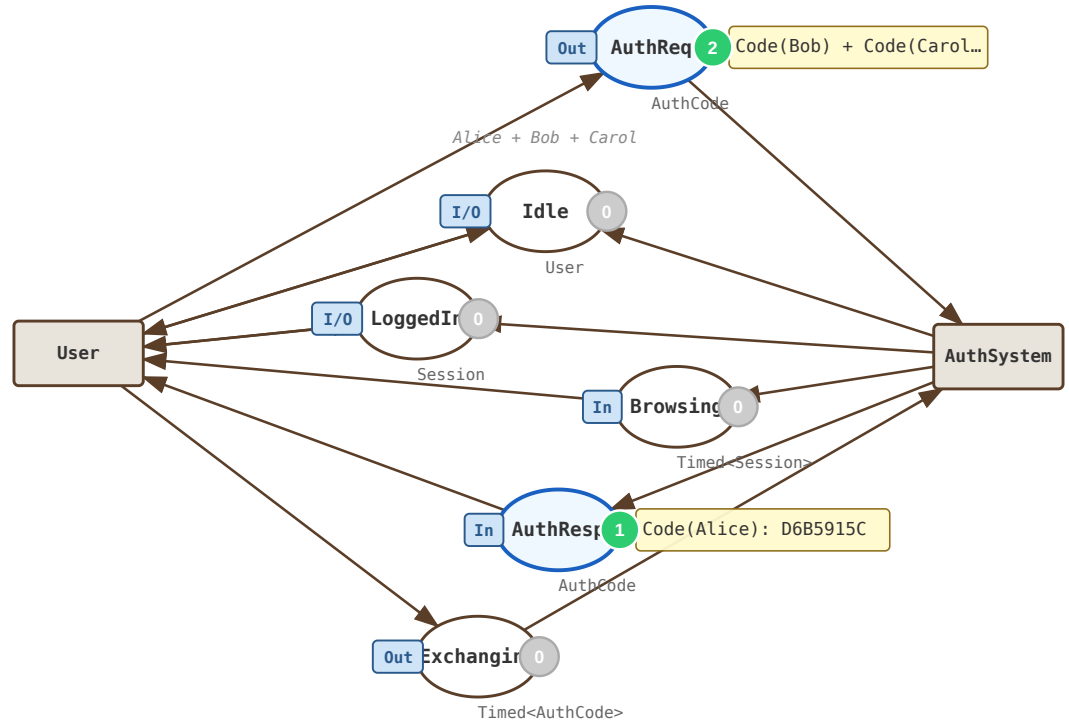

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- Two sub-pages: User & AuthSystem

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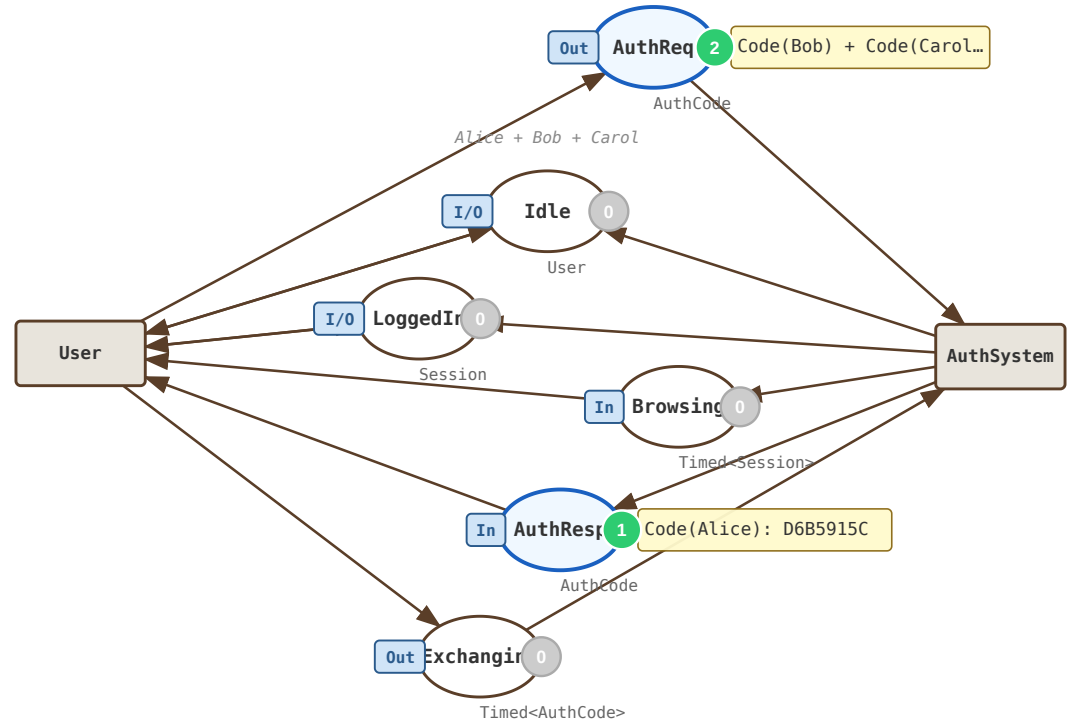

Colour sets are just C# types

Any `notnull, IEquatable<T>` type works — primitives, enums, records, classes.

```
public record User(int Id, string Name) {
  public string CsrfState  => $"s{Id}";
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public record Session(User User, int TokenId);
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Places & multisets

```
// inside the model constructor
var authReq = AddPlace<AuthCode>("AuthReq");            // empty initial marking

var idle    = AddPlace("Idle", Multiset.Of(
                new User(1,"Alice"), new User(2,"Bob"), new User(3,"Carol")));

var exchg   = AddTimedPlace<AuthCode>("Exchanging");    // timed place
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Transitions: a fluent builder

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var u   = new Var<User>("u");

AddTransition("Login")
  .Input(idle, u).Output(waiting, u)
  .Output(authReq, () ⇒ AuthCode.Create(u.Val), "AuthReq")
  .Build();

AddTransition("ReceiveCode")
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Timed CPNs

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AddTransition("GrantToken")
  .TimedInput(exchg, c).Input(nextTokId, tid)
  .Output(nextTokId, () ⇒ tid.Val + 1)
  .Output(loggedIn,  () ⇒ new Session(c.Val.User, tid.Val))
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AddTransition("SessionExpire")
  .TimedInput(expiry, se).Input(loggedIn, se2)
  .Guard(() ⇒ se.Val.TokenId == se2.Val.TokenId)
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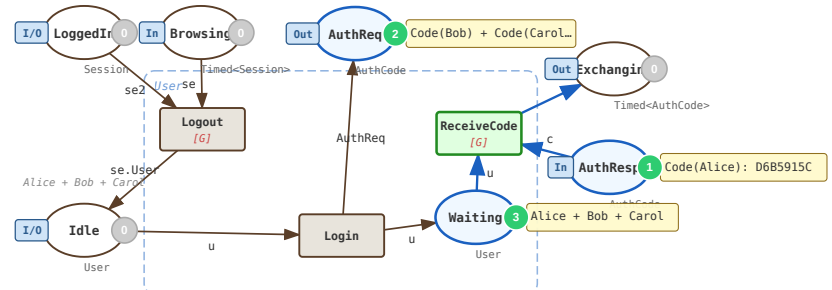
Hierarchical CPNs

```
class UserPage : CpnPage {
  public UserPage(Place<User> idle,
       Place<AuthCode> authReq, ...)
    : base("User") {
    InOut(idle);
    Out(authReq); In(authResp);
    Out(exchg);
    InOut(loggedIn); In(browsing);
    var waiting = AddPlace<User>("Waiting");
    // Login, ReceiveCode, Logout
  }
}
```




```
AddSubPage(new UserPage(...),       "User");
AddSubPage(new AuthSystemPage(...), "AuthSystem");
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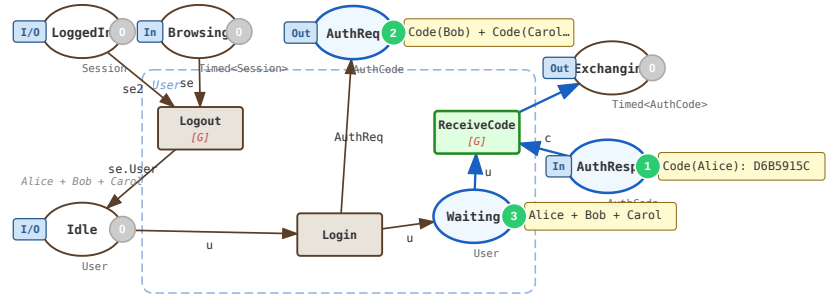

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```

Hierarchical CPNs

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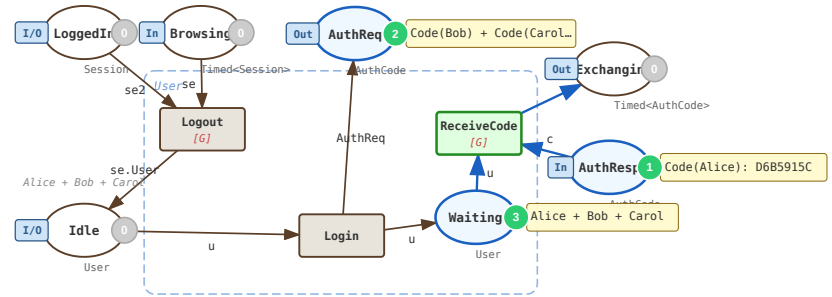

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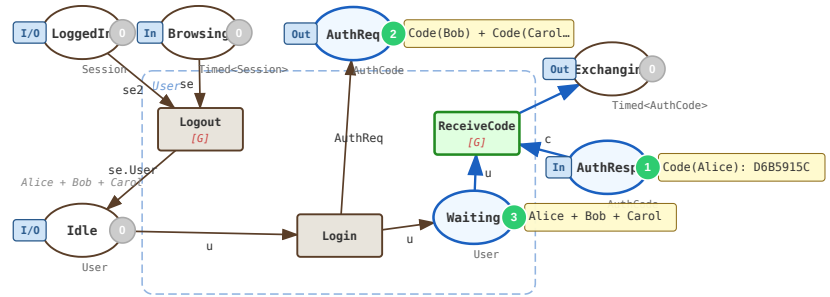

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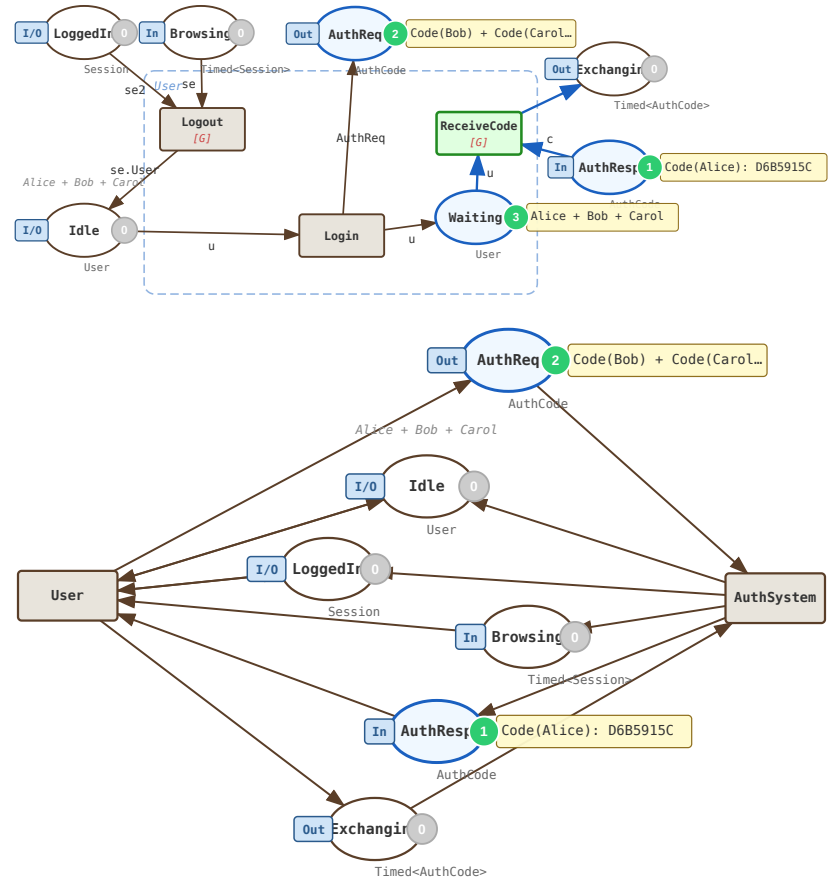

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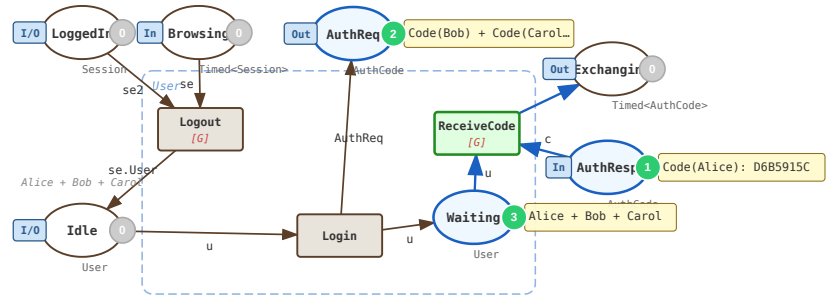
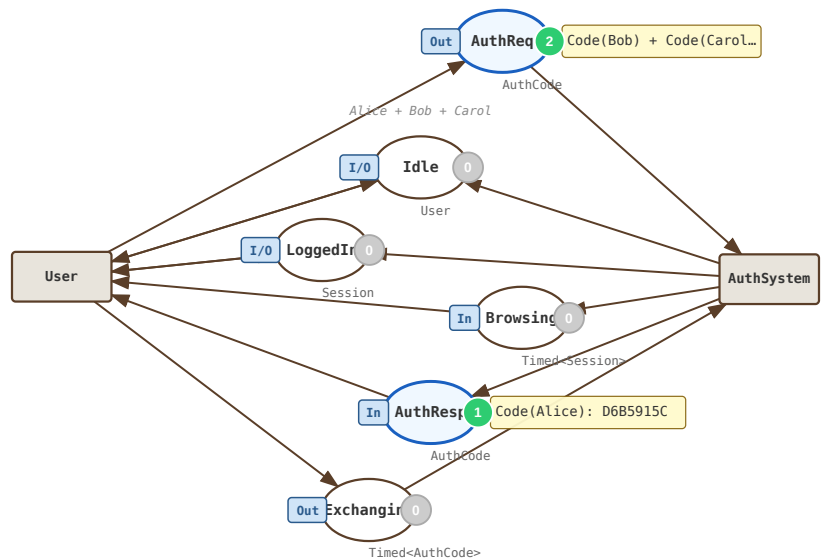
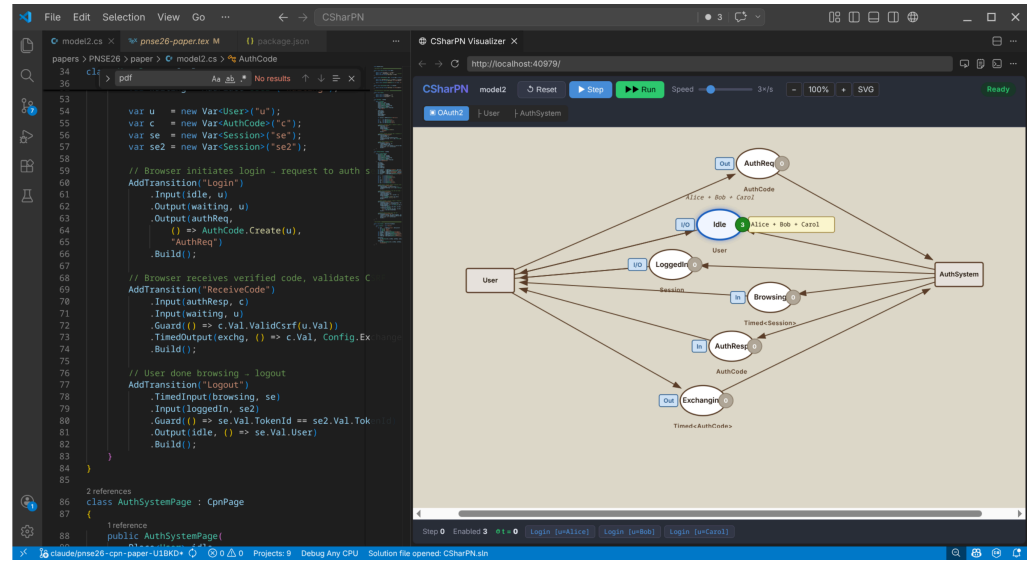




Visualiser & VS Code integration



Visualiser & VS Code integration

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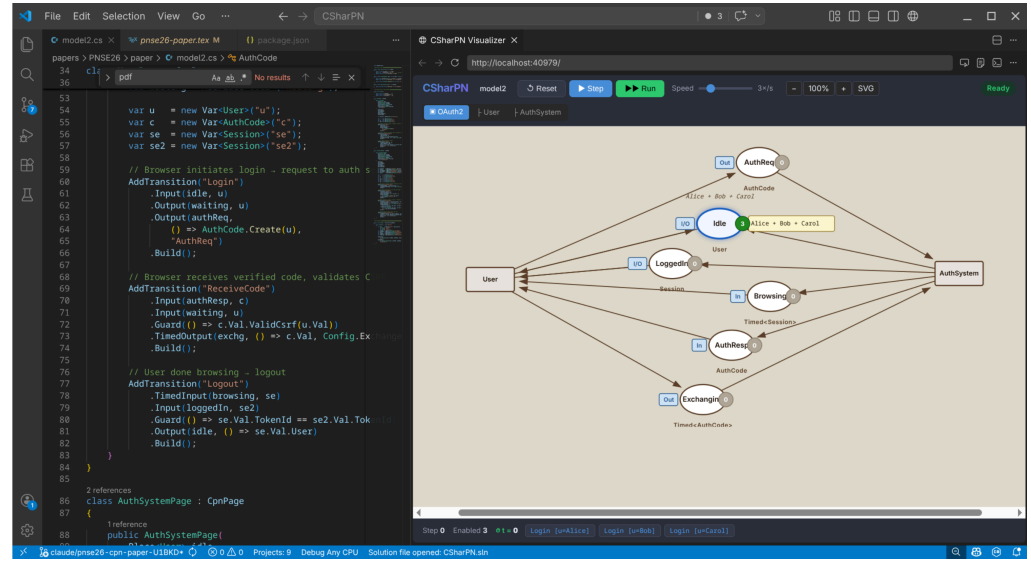

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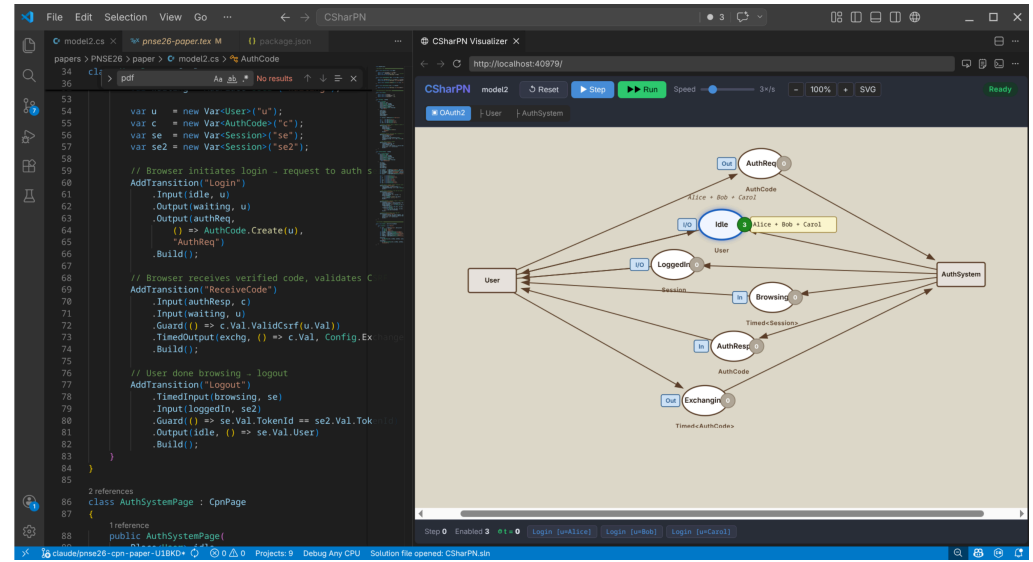

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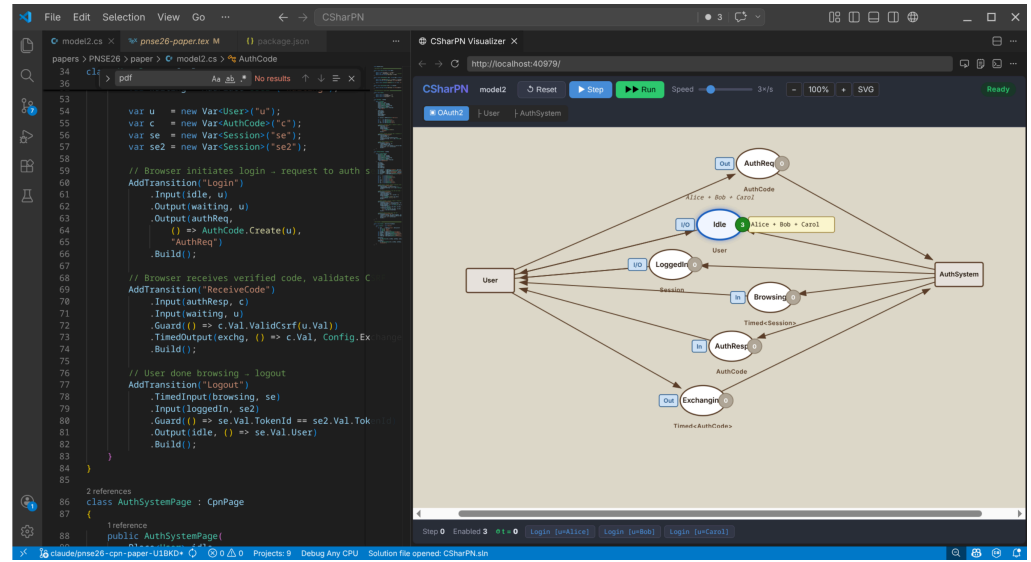

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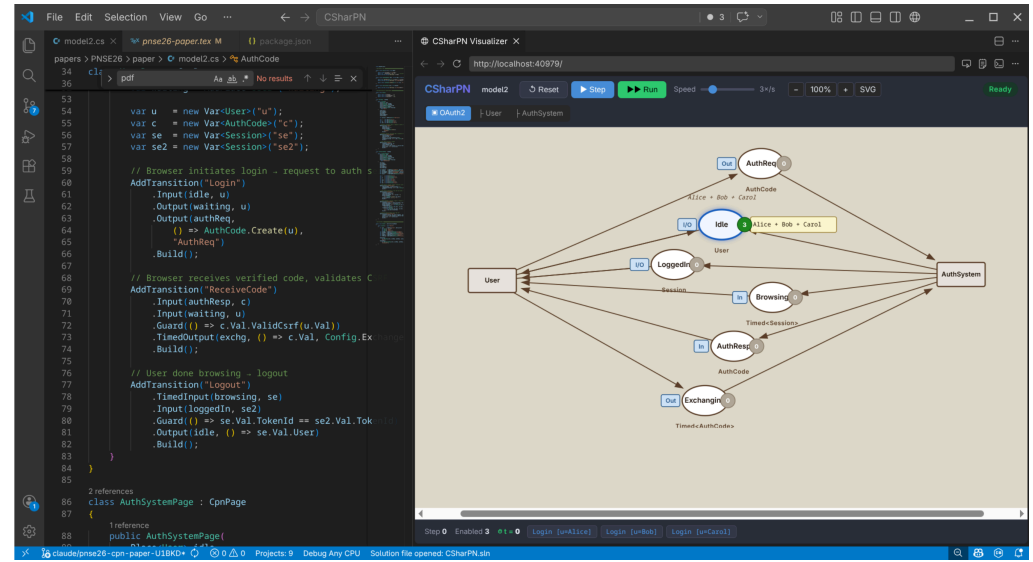

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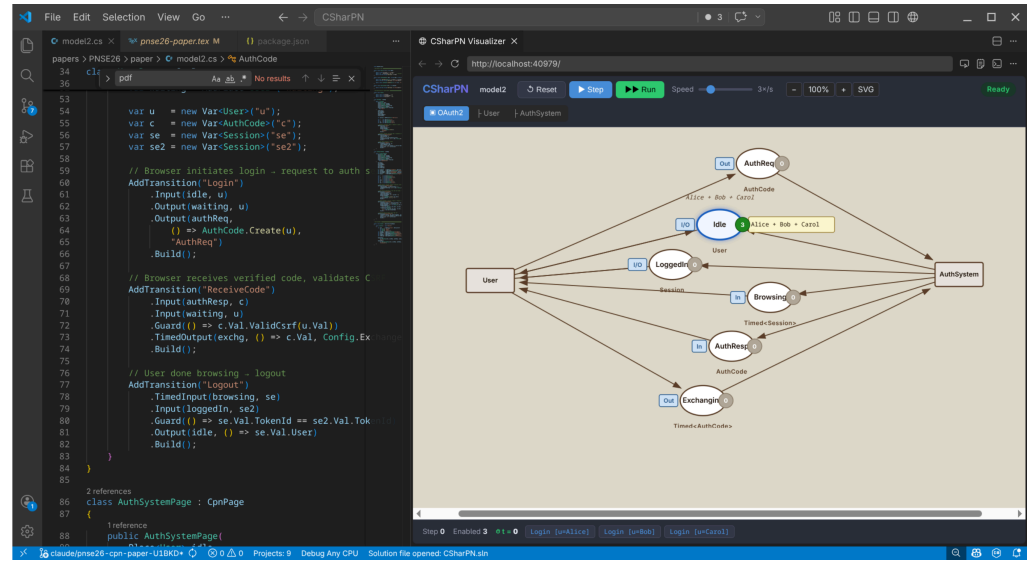

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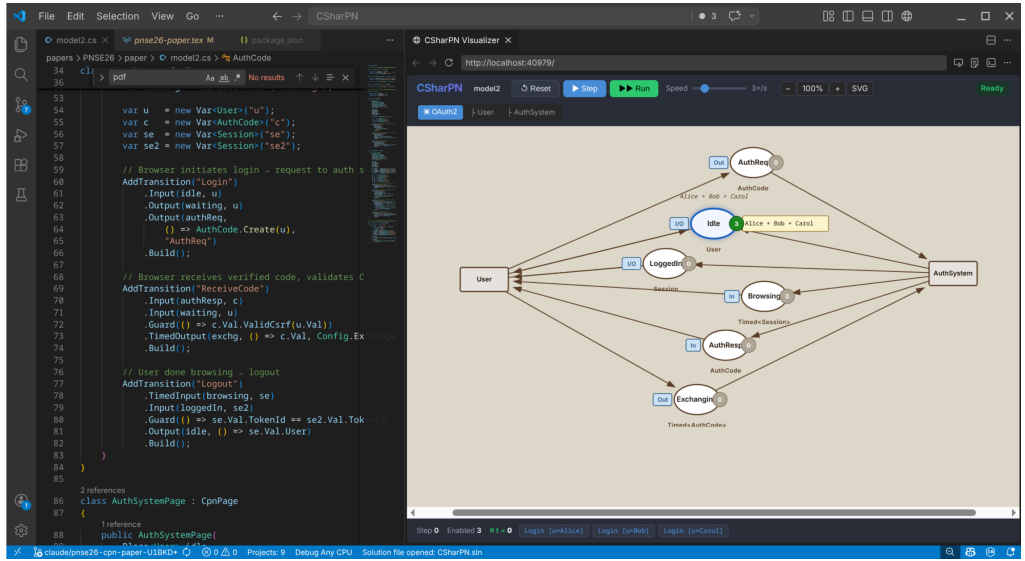

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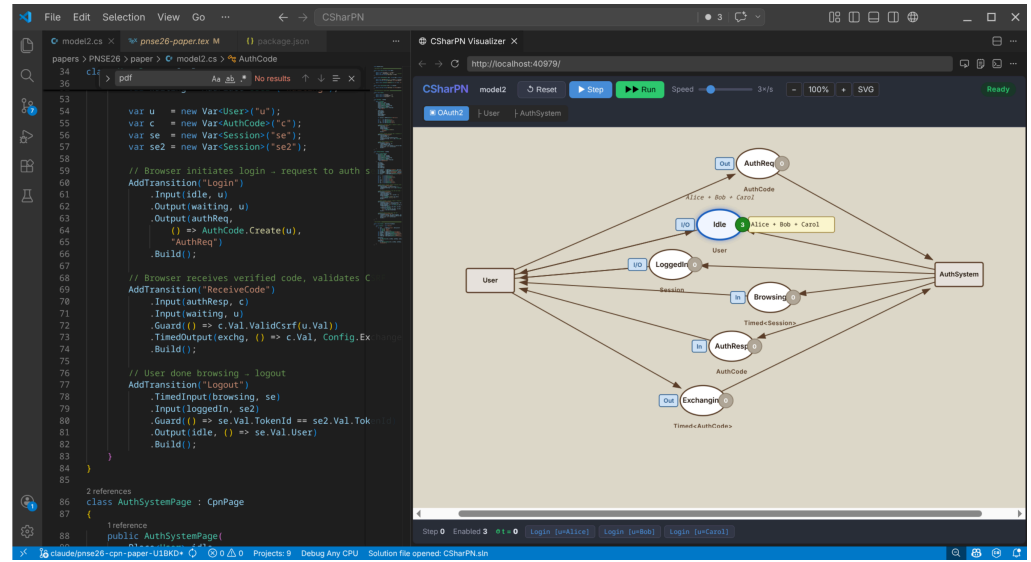

Future work

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- State-space analysis on `CpnState` snapshots

Thank you

Questions?

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`github.com/simontjell/CSharPN.public`

Simon Tjell · simon@simplesystemer.dk